

2016: selected excess 1 at a 91 MeV signal hypothesis

Neighboring signal-MC template; window $-3.5u$ to $+3.5u$ | $c = 90.731$ MeV, $s = 3.492$ MeV

Signed full-selected yield = $20,201 \pm 5,699$; local asymptotic $Z = 3.55 \sigma$, $p = 0.000196$

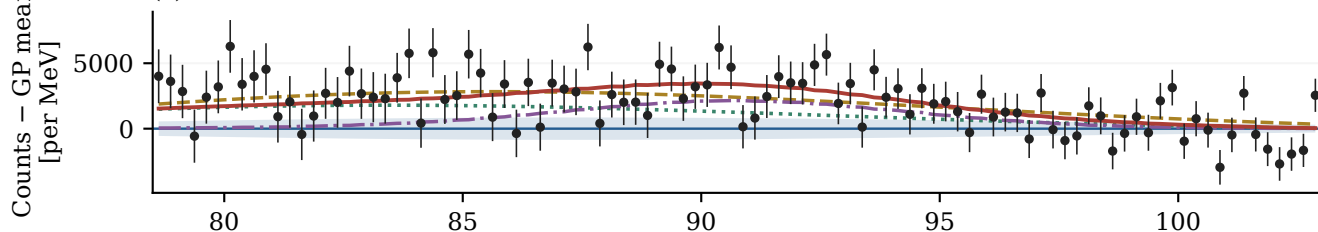
Fixed-source null toys: local rank $p = 0.004$ (3 / 1000 exceedances); count 95% interval [0.000619, 0.00874]

- Observed counts
- ⋯ Background in signed S+B fit
- ⬜ Fit interval: excluded from GP training
- ▒ GP mean and ± 1 SD constraint
- Signed signal + background
- Search mass hypothesis
- Background-only fit
- Signed signal (middle panel)
- Signal core center

(a) Spectrum in the fit interval



(b) Residuals relative to the GP mean



(c) Standardized residuals and excluded fit interval

